

Ahmad Shabani (Ph.D.)

Assistant Professor
Amirkabir University of Technology



 a.shabani@aut.ac.ir

 linkedin.com/in/ahmad-shabani-316460101/

 <https://scholar.google.com/citations?user=iU2Rx7UAAAAJ>

 <https://aut.ac.ir/>

 +767600953073

 0000-0002-4690-4430

 ahmadshabani89@gmail.com

 Hammarbyvägen 68, 19436
Upplands Väsby.

PROFILE

Ahmad Shabani is a researcher in hardware and computer security domain, and digital ASIC design. He is an Assistant Professor at Amirkabir University of Technology and has recently joined KTH University as a researcher. Ahmad's research primarily focuses on developing efficient countermeasures against Hardware Trojans and lightweight PUF-based authentication protocols. What sets his research apart is its focus on the hardware security at different abstraction levels from the microarchitectural-level down to the physical level. His research and academic contributions have been recognized through numerous publications in prestigious journals, as well as by receiving esteemed awards. He is also proficient in digital Integrated Circuit (ASIC) design, covering both front-end, verification, and back-end design stages of the digital system and processors. His research interests include security analysis of the transient microarchitectural attack, design for hardware trust, and side-channel aware implementation of the crypto primitives.

EDUCATION

Nov 2025 – present
Stockholm, Sweden

PostDoc Researcher: Hardware and Computer Security
KTH Royal Institute of Technology 

The research focuses on improving the security of computer processors. Modern processors are very complex and can have hidden vulnerabilities that hackers can exploit to steal confidential data from computers.

Apr 2016 – Aug 2021
Tehran, Iran

Ph.D.: Digital Electronic Systems
University of Tehran 

GPA: (18.02 Out of 20)

Ph.D. Dissertation: "Hardware Trojan Detection Approaches Using Design for Hardware Trust"; (Score: **Excellent**)

Goal: Design efficient methodologies based on Design for Hardware Trust (DfTr) to detect hardware Trojans mounted by untrusted foundry.

2014 – 2016
Tehran, Iran

MSc: Digital Electronic Systems
Shahid Beheshti University 

GPA: (19.37 Out of 20)

MSc Dissertation: "Design and Implementation of Low-Power Discrete Cosine Transform Using CORDIC Algorithm"; (Score: **Excellent**).

Goal: Design and implement a low-power DCT architecture based on CORDIC algorithm for Wireless Capsule Endoscopy and compressor engine in emerging High Efficiency Video Coding (HEVC).

WORK EXPERIENCE

Dec 2024 – Present

Assistant Professor (on Sabbatical Leave)

Amirkabir University of Technology (Tehran Polytechnic) 

Course I: "Computer Architecture RISC-V Edition"

Course II: "Introduction of Logic Design"

Course III: "Cyber-Physical Systems"

Course IV: "Digital System Design Automation Using VHDL"

Dec 2023 – Oct 2025
Tehran, Iran

Embedded Linux Software Dev. (Part-time)

CROUSE PJS Co. 

- FPGA-Based Embedded System Design for Validation and Test
- Custom PetaLinux Build for Zynq-7000 SoCs and PYNQ Integration
- Linux Driver Development for General and Custom Xilinx IP Core
- Hardware/Software Co-Design and System Integration

Apr 2023 – Dec 2023
Tehran, Iran

Automotive Engineer (Project-Based)

Azin Electro Idea. 

- Working on the communication protocols and security measures in Passive-Entry Passive Start (PEPS) system.
- Working on the DPA analysis for breaking KeeLoQ protocol available in Key-Fobs.

Mar 2021 – Apr 2023
Tehran, Iran

Digital ASIC Designer

Mana Excellence Center (Full-time)

- Manager of Digital ASIC Design of Automotive-Graded ICs;
- Participation in synthesis, simulation, LEC, coverage estimation, and static timing analysis (STA);
- Participation in physical design including partitioning, floorplanning, power planning, placement, CTS, and routing;

2020 – Oct 2025
Tehran, Iran

CTO & Technical Consultant (Full-time until 2021, then part-time)

Shahab Co. 

Head of the electronic department working on:

- Design non-destructive & low-cost smart electricity meters based on IoT including the hardware and firmware parts;
- Developing small-sized devices including hardware and firmware to monitor non-smart gas meters using computer vision and image processing.

HONORS & AWARDS

Recipient of Prestigious Award for Outstanding Ph.D. Dissertation.

19th International ISC Conference on Information Security and Cryptography (ISCISC2022), 2022

Ranked 1st Among Graduate Students in Master's Degree Program.

Shahid Beheshti University, Digital Electronic Systems, Tehran, IRAN, 2016.

Recipient of Student Prize from Iran's National Elites Foundation.

Recipient of Student Prize for two successive years, Iran's National Elites Foundation, 2017-2018.

Admitted to Ph.D. Degree Program with Exceptional Merit, Waiving Entrance Exam Requirement.

Organization for Development of Exceptional Talents, University of Tehran, Tehran, Iran, 2016.

Guest Speaker at the International ISC Conference on Information Security and Cryptology (ISCISC2023)

Presenting a speech titled "Empowering Hardware Security Through CAD Tools for Hardware Trojan Detection" Iranian Society of Cryptology, 2023.

SKILLS

Hardware Description Language (VHDL/Verilog)



Proficient in synthesizable hardware description

Mentor Graphics QuestaSim



efficient simulation tool for hardware description

Calibre (DRC, LVS, PEX)



Post layout checking and netlist extraction

Sensors & Modules



Accelerator (MPU6050), OLED (1.3", 0.96"), Hall Effect (44E, AK09911C), IR, PIR (HC-SR501), ADC (MCP3008, ADS1115), RTC (PCF8563, PCF8523), WiFi (ESP-07, ESP-12E, ESP-12N), Light (GY-302)

Development Boards



ESP32CAM, ESP32 (WROOM-32), Arduino Uno/Mega, NodeMCU, Raspberry Pi (ZeroW), Raspberry Pi Pico.

Programming Language



Perl, C/C++, Python, Fortran, CUDA C, SystemC, TCL, VHDL, Verilog.

Synopsys IC Design Toolchain



Synopsys Design Compiler, Synopsys Formality, Synopsys VCS, Synopsys DFT, Synopsys PrimeTime.

Cadence IC Design Toolchain



Virtuoso, Innovus, Voltus, Quntus, Abstract Generator.

Communications Protocols



SPI, I2C, UART, CAN, LIN, RS485, RS232, MQTT, HTTP/HTTPS, WiFi, BLE, Ethernet.

EDA & IDE



Altium Designer, Icarus Verilog, Quartus(Intel), Vivado Design Suite, Vitis, Petalinux, Netbeans, VSCode, Atmel Studio.

Microcontroller & SOCs



ESP32, ESP8266, AVR (ATmega 8, 16, 32, 328), ATtiny85, STM32F103C8T6, NXP-LPC1788

FPGA-Based Embedded System Design



ZYNQ Embedded Platform, ZTurn, Alinx, Zybo.

PROJECTS (ASIC, HW, SW)

Design Automatic Chip Tester for Automotive Application Based on Zynq FPGA SOC.

CROUSE PJS Co. 2024

An automatic parametric and functional IC tester using Zynq 7020 SOC Design working on both PL and PS sides of the Zynq. The target IC is high-speed and fault-tolerant CAN bus transceivers.

"Digital ASIC Design (Front-End & Back-End) of High-Speed and Fault-Tolerant CAN Transceivers" Mana Excellence Center, 2022-2023.

Designing the digital part of high-speed and fault-tolerant CAN transceiver IC.

Digital ASIC Design (Verification and Back-End) of Immobilizer IC Used in Automotive Key-Fobs, Freelance Project, 2023.

Designing the digital part of Transponder IC available in automotive Key-Fobs

"Digital ASIC Design (Front-End & Back-End) of LIN Transceiver"

Mana Excellence Center, 2022-2023.

Designing the digital part of LIN transceiver.

"Smart Electricity Meter (Hardware & Firmware Development)"

Shahab-Co, 2021-2022.

Low-cost non-destructive ESP-based smart electricity meter able to measure energy, voltage, and current of twelve input channels, and send them to the server in real-time using WiFi. For more info visits (<https://github.com/hiddenman23?tab=repositories>).

"Gas Meter Telemetry Add-on Based on Computer-Vision"

Shahab-Co, 2021-2022.

A small-sized device capable of adding telemetry and supervised monitoring to the traditional non-smart gas meters. The device incorporates a low-cost camera and can process the image taken from the gas meter, extract the digits (locally or server-side), and send the result to the server via WiFi.

SELECTED PUBLICATIONS (SEE SCHOLAR PROFILE)

A Sajadi, A Shabani, B Alizadeh; "DC-PUF: Machine learning-resistant PUF-based authentication protocol using dependency chain for resource-constraint IoT devices", Journal of Network and Computer Applications, 2023.

Khormizi, Fatemeh, Ahmad Shabani, and Bijan Alizadeh. "Hardware Patching Methodology for Neutralizing Timing Hardware Trojans Using Vulnerability Analysis and Time Borrowing Scheme." IEEE Transactions on Circuits and Systems II: Express Briefs (2022).

Ashtari, Amir, and Bijan Alizadeh. "A comparative study of machine learning classifiers for secure RF-PUF-based authentication in internet of things." Microprocessors and Microsystems 93 (2022): 104600.

M Sabri, A Shabani, B Alizadeh - "SAT-Based Integrated Hardware Trojan Detection and Localization Approach Through Path-Delay Analysis", IEEE Transactions on Circuits and Systems II: Express, 2021

Shabani, Ahmad, and Bijan Alizadeh. "Enhancing hardware Trojan detection sensitivity using partition-based shuffling scheme." IEEE Transactions on Circuits and Systems II: Express Briefs 68.1 (2020): 266-270.

Shabani, Ahmad, and Bijan Alizadeh. "PODEM: A low-cost property-based design modification for detecting hardware trojans in resource-constraint IoT devices." Journal of Network and Computer Applications 167 (2020): 102713.

Shabani, A., Sabri, M., Khabbazan, B., & Timarchi, S. (2020). "Area and Power-Efficient Variable-Sized DCT Architecture for HEVC Using Muxed-MCM Problem". IEEE Transactions on Circuits and Systems I: Regular Papers, 68(3), 1259-1268.

A. Shabani, B. Alizadeh; "PMTP: A MAX-SAT Based Approach to Detect Hardware Trojan Using Propagation of Maximum Transition Probability", Journal of Transaction of Computer Aided Devices (TCAD), 2019.

RESEARCH INTERESTS

- Hardware Trojans & Security
- Design for Hardware Trust
- Transient Microarchitectural Attacks
- PUF-Based Authentication
- Low-Power VLSI Design

REFERENCES

Dr. Somayeh Timarchi,
School of Electrical Engineering and Computer Science, Queen Mary University of London,
Email: s.timarchi@qmul.ac.uk

Dr. Bijan Alizadeh, Associate Professor, Department of Electronics, University of Tehran, Tehran, Iran.
Email: b.alizadh@ut.ac.ir

Dr. Saeed Saiedi, Faculty of Electrical and Computer Engineering, Tarbiat Modares University, Tehran, Iran.
Email: saeed.saeedi@modares.ac.ir

Dr. Hamed Dehdashti,
Associate Professor, Department of Electrical and Computer Engineering, Jahrom University, Jahrom, Iran.
Email: jahromi@jahromu.ac.ir